Label Smoothing under Noisy Labels

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Abstract & Motivation

- Label Smoothing (LS): mixes hard labels with a uniform label to regularize and improve generalization.
- At high label-noise rates, LS can oversmooth and hurt performance.
- Negative/Not Label Smoothing (NLS): uses negative smoothing to counter oversmoothing; more robust at high noise.
- There is a phase transition of the optimal smoothing rate; NLS preferred when noise is high.
- Experiments across benchmarks confirm; links to loss correction, NLNL, and Peer Loss.

Figure 1 placeholder: Optimal smooth rates vs. noise (UCI datasets)

Label Smoothing vs Negative Label Smoothing

LS
- Creates a soft target by mixing the one-hot label with a uniform vector.
- Reduces overconfidence; aids calibration.
- Helps under moderate noise, but may oversmooth at high noise.

NLS
- Uses a negative smoothing rate: increases target weight, subtracts from others.
- Not a probability vector but valid for losses linear in labels (e.g., cross-entropy).
- Boosts confidence on correct predictions; penalizes wrong classes.

Generalized Label Smoothing (GLS) unifies LS and NLS by allowing negative smoothing.

Learning with Noisy Labels

Provides ground-truth supervision and feature distribution for defining the target classifier on clean data.

1 Clean data (X, Y)
Input: features X with clean labels Y from D.
Output: pairs (X,Y) before corruption.

1 estimate →

Encodes flip rates; estimated or known; drives label corruption.

2 Transition matrix T
Obtained as given in synthetic setups or estimated from a clean subset/confusion stats; it specifies P(Ỹ=j|Y=i) and determines corruption severity.

2 corrupt →

Ỹ is sampled by applying T; needed to simulate/reflect imperfect supervision during training.

3 Noisy labels (X, Ỹ)
Input: (X,Y) and T.
Output: observed training pairs (X,Ỹ) drawn from ˜D.

3 smooth ↓↓

Choose smoothing rate r (can be negative). r<0 increases the target weight and subtracts from non-targets; r≥0 spreads mass uniformly.

4 Apply GLS on Ỹ
Input: one-hot Ỹ and smoothing rate r.
Output: smoothed targets $\tilde{Y}_{GLS,r} = (1-r)\cdot\tilde{Y} + r/K\cdot 1$ (r can be negative).

4 optimize ↓↓

5 Train model f(X)
Input: $(X, \tilde{Y}_{GLS,r})$.
Output: classifier f optimized with cross-entropy on smoothed noisy targets.

Optimize cross-entropy on $\tilde{Y}_{GLS,r}$; learns probabilities over classes via softmax and adjusts confidence (LS lowers, NLS raises).

Model Confidence

- Confidence compares the predicted probability of the target class to others.
- With cross-entropy, higher confidence corresponds to lower loss on the true class.
- LS generally decreases confidence; NLS increases it.

Illustration placeholder: confidence scale vs smoothing

Risk Minimization with GLS under Noise

- Noisy GLS risk equals the clean target risk plus two bias terms that alter confidence.
- There is a phase transition: the best smoothing depends on noise level and the clean-data optimum.
- Goal: choose smoothing to recover the classifier that is optimal on clean data.

Block diagram: Noisy risk → (Clean risk + Bias1 + Bias2)

Symmetric Noise: Optimal Smoothing

- There is a threshold where the confidence bias cancels.
- If noise is below the threshold: prefer LS.
- At the threshold: vanilla training (no smoothing) is optimal.
- Above the threshold: prefer NLS.

Figure 2 placeholder: Decision regions for LS/NLS vs noise

NLS is favored as noise increases and when the clean-data optimal smoothing is small.

Asymmetric Noise: Confidence Imbalance

- Under asymmetric flips, one bias remains and skews confidence across classes.
- The model becomes overconfident on the lower-noise class and less confident on the higher-noise class.

Graphic placeholder: confidence shift per class under asymmetry

Empirical Risk and Generalization

- Generalization bounds connect expected noisy GLS risk to empirical risk.
- Practice: warm up with cross-entropy, then switch to NLS to solidify patterns learned from clean/easy samples.
- High noise already smooths supervision; NLS counters oversmoothing.

Multi-class Extensions

- Confidence extends by contrasting the true class against the average of others.
- Sparse noise: decomposes into binary pairs; the same LS/NLS rule applies.
- Symmetric multi-class noise: the optimal smoothing has a closed form balancing noise and the clean optimum.
- Decision rule mirrors binary: low noise → LS; high noise → NLS.

Empirical Takeaways (Consolidated)

- Across UCI, CIFAR-10/100, and real-world noisy labels, the optimal smoothing shifts from positive to negative as noise increases.
- Low noise: vanilla or LS performs best; moderate to high noise: NLS outperforms LS.
- NLS raises confidence on correct predictions and lowers it on wrong ones, improving separability.
- Warm-up with cross-entropy then switching to NLS is effective in practice.

Summary graphic placeholder: noise level → preferred smoothing (LS → NLS)

Connections: Loss Correction

- Backward/Forward loss correction align with NLS under suitable smoothing and add a small bias term.
- Under symmetric noise, loss correction reduces exactly to a special case of NLS.

Mapping diagram placeholder: LC ↔ NLS

Connections: Complementary Labels

- Training with complementary labels (penalizing the non-target) corresponds to the limit of NLS with very negative smoothing.

Formula placeholder: CL → NLS limit

Connections: Peer Loss

- Peer loss behaves like NLS with an additional bias term that diminishes when class priors are balanced.
- With equal noisy class priors, peer loss matches an exact NLS form.

Mapping placeholder: Peer Loss ↔ NLS

Practical Guidance

- Low noise: prefer LS or no smoothing.
- High noise: switch to NLS (optionally after cross-entropy warm-up).
- Monitor confidence: NLS increases separation between correct and wrong predictions.
- Combine NLS with robust training methods for further gains.

Conclusions

- Optimal smoothing is noise-dependent with a clear transition from LS to NLS as noise grows.
- NLS combats oversmoothing by adjusting confidence appropriately under noisy labels.
- Connections to loss correction, complementary labels, and peer loss support the theory.
- Extensive experiments validate the findings.

Future: explore negative labels beyond classification and broader architectures.

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